Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

IT13: ADBMS

Chapter 2:SQL

Practice Assignment Solution Template

Write a query and paste a screenshot of your output below the question.

1. Create a database named CompanyDB.

   Query: CREATE DATABASE CompanyDB;

```
mysql> CREATE DATABASE CompanyDB;
Query OK, 1 row affected (0.013 sec)

mysql>
```

2. Create a table Employees with the following columns:

   Employees (EmpID (Integer, Primary Key), EmpName, Salary, JoinDate)

   Query: CREATE TABLE Employees (EmpID int primary key, EmpName varchar(50), Salary float, JoinDate date);

```
mysql>
mysql> CREATE TABLE Employees (EmpID int primary key, EmpName varchar(50), Salary float, JoinDate date);
Query OK, 0 rows affected (0.032 sec)
```

3. Create a table Departments with:

   Departments (DeptID (Integer, Primary Key), DeptName)

   Query: CREATE TABLE Departments (DeptID int primary key, DeptName varchar(30));

```
mysql> CREATE TABLE Departments (DeptID int primary key, DeptName varchar(30));
Query OK, 0 rows affected (0.015 sec)
```

4. Add a column Email (VARCHAR(100)) to the Employees table.

   Query: ALTER TABLE Employees ADD COLUMN (Email varchar(100));

```
mysql> ALTER TABLE Employees ADD COLUMN (Email varchar(100));
Query OK, 0 rows affected (0.043 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

5. Add a column dept_location to the Departments table.

   Query: ALTER TABLE Departments ADD COLUMN (dept_location varchar(100));

```
mysql> ALTER TABLE Departments ADD COLUMN (dept_location varchar(100));
Query OK, 0 rows affected (0.039 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

6. Add a NOT NULL Constraint to the Phone column

   Query: ALTER TABLE Employees ADD COLUMN Phone varchar(15); ALTER TABLE Employees MODIFY COLUMN Phone varchar(15) NOT NULL;

```
mysql> ALTER TABLE Employees ADD COLUMN Phone varchar(15);
Query OK, 0 rows affected (0.043 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employees;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(50)  | YES  |     | NULL    |       |
| Salary   | float        | YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
| Email    | varchar(100) | YES  |     | NULL    |       |
| Phone    | varchar(15)  | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
6 rows in set (0.003 sec)

mysql> ALTER TABLE Employees MODIFY COLUMN Phone varchar(15) NOT NULL;
Query OK, 0 rows affected (0.037 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employees;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(50)  | YES  |     | NULL    |       |
| Salary   | float        | YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
| Email    | varchar(100) | YES  |     | NULL    |       |
| Phone    | varchar(15)  | NO   |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
6 rows in set (0.001 sec)
```

7. Change the size of the EmpName column from VARCHAR (50) to VARCHAR (100).

   Query: ALTER TABLE Employees MODIFY COLUMN EmpName varchar(100);

```
mysql> ALTER TABLE Employees MODIFY COLUMN EmpName varchar(100);
Query OK, 0 rows affected (0.046 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employees;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(100) | YES  |     | NULL    |       |
| Salary   | float        | YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
| Email    | varchar(100) | YES  |     | NULL    |       |
| Phone    | varchar(15)  | NO   |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
6 rows in set (0.003 sec)
```

8. Drop the primary key Constraint from Departments

   Query: ALTER TABLE Departments DROP PRIMARY KEY;

```
mysql>
mysql> ALTER TABLE Departments  DROP PRIMARY KEY;
Query OK, 0 rows affected (0.050 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Departments
    -> ;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| DeptID        | int          | NO   |     | NULL    |       |
| DeptName      | varchar(30)  | YES  |     | NULL    |       |
| dept_location | varchar(100) | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
3 rows in set (0.002 sec)
```

9. Create a foreign key constraint between two tables

   Query:

   1. ALTER TABLE Departments ADD PRIMARY KEY (DeptID);
   2. ALTER TABLE Employees ADD COLUMN DeptID int REFERENCES Departments(DeptID);

```
mysql> ALTER TABLE Departments ADD PRIMARY KEY (DeptID);
Query OK, 0 rows affected (0.037 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE Employees ADD COLUMN DeptID int REFERENCES Departments(DeptID);
Query OK, 0 rows affected (0.040 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employees;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| EmpID    | int          | NO   | PRI | NULL    |       |
| EmpName  | varchar(100) | YES  |     | NULL    |       |
| Salary   | float        | YES  |     | NULL    |       |
| JoinDate | date         | YES  |     | NULL    |       |
| Email    | varchar(100) | YES  |     | NULL    |       |
| Phone    | varchar(15)  | NO   |     | NULL    |       |
| DeptID   | int          | YES  | MUL | NULL    |       |
+----------+--------------+------+-----+---------+-------+
7 rows in set (0.002 sec)
```

10. Rename the column EmpName to EmployeeName.

    Query: ALTER TABLE Employees RENAME COLUMN EmpName TO EmployeeName;

```
mysql> ALTER TABLE Employees RENAME COLUMN EmpName TO EmployeeName;
Query OK, 0 rows affected (0.025 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employees;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| EmpID        | int          | NO   | PRI | NULL    |       |
| EmployeeName | varchar(100) | YES  |     | NULL    |       |
| Salary       | float        | YES  |     | NULL    |       |
| JoinDate     | date         | YES  |     | NULL    |       |
| Email        | varchar(100) | YES  |     | NULL    |       |
| Phone        | varchar(15)  | NO   |     | NULL    |       |
| DeptID       | int          | YES  | MUL | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
7 rows in set (0.003 sec)

mysql>
```

11. Apply a CHECK constraint on the Age and Salary columns.

    Query:

    1. ALTER TABLE Employees ADD COLUMN Age int;
    2. ALTER TABLE Employees ADD CONSTRAINT check_age CHECK (AGE > 18);
    3. ALTER TABLE Employees ADD CONSTRAINT check_salary CHECK (Salary > 0);

```
mysql> ALTER TABLE Employees ADD COLUMN Age int;
Query OK, 0 rows affected (0.037 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE Employees ADD CONSTRAINT check_age CHECK (AGE > 18);
Query OK, 0 rows affected (0.156 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE Employees ADD CONSTRAINT check_salary CHECK (Salary > 0);
Query OK, 0 rows affected (0.054 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

12. Remove the Phone column from the Employees table.

Query: ALTER TABLE Employees DROP COLUMN Phone;

```
mysql> ALTER TABLE Employees DROP COLUMN Phone;
Query OK, 0 rows affected (0.048 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Employees;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| EmpID        | int          | NO   | PRI | NULL    |       |
| EmployeeName | varchar(100) | YES  |     | NULL    |       |
| Salary       | float        | YES  |     | NULL    |       |
| JoinDate     | date         | YES  |     | NULL    |       |
| Email        | varchar(100) | YES  |     | NULL    |       |
| DeptID       | int          | YES  | MUL | NULL    |       |
| Age          | int          | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
7 rows in set (0.002 sec)
```

13. Rename the table Employees to EmployeeDetails.

Query: RENAME TABLE Employees To EmployeeDetails;

```
mysql> RENAME TABLE Employees To EmployeeDetails;
Query OK, 0 rows affected (0.014 sec)

mysql> DESC Employees;
ERROR 1146 (42S02): Table 'CompanyDB.Employees' doesn't exist
mysql> DESC EmployeeDetails;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| EmpID        | int          | NO   | PRI | NULL    |       |
| EmployeeName | varchar(100) | YES  |     | NULL    |       |
| Salary       | float        | YES  |     | NULL    |       |
| JoinDate     | date         | YES  |     | NULL    |       |
| Email        | varchar(100) | YES  |     | NULL    |       |
| DeptID       | int          | YES  | MUL | NULL    |       |
| Age          | int          | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
7 rows in set (0.001 sec)
```

14. Drop the CHECK constraints.

Query: ALTER TABLE EmployeeDetails DROP CHECK check_salary;

```
mysql> ALTER TABLE EmployeeDetails DROP CHECK check_salary;
Query OK, 0 rows affected (0.020 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE EmployeeDetails DROP CHECK check_age;
Query OK, 0 rows affected (0.011 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
```

15. Delete the Departments table permanently.

Query:

1. ALTER TABLE EmployeeDetails DROP FOREIGN KEY EmployeeDetails_ibfk_1;
2. DROP TABLE Departments;

```
mysql> ALTER TABLE EmployeeDetails DROP FOREIGN KEY EmployeeDetails_ibfk_1;
Query OK, 0 rows affected (0.021 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DROP TABLE Departmentsl
    -> ^C
mysql> DROP TABLE Departments;
Query OK, 0 rows affected (0.021 sec)

mysql>
```

16. Drop the database.

Query: DROP DATABASE CompanyDB;

```
mysql> DROP DATABASE CompanyDB;
Query OK, 1 row affected (0.026 sec)

mysql>
```